

GAIRA V7 — Phase 08

Hierarchical molecular retrieval

Can chemistry-aware retrieval beat direct CSM retrieval, while remaining fully explainable?

Status	COMPLETE — outcome A
Decision	keep direct CSM retrieval
Baselines reproduced	EXACT
BSV2 on the inference path	NO — not imported anywhere in the package
A raw spectrum top-1 / MRR	0.6347 / 0.6919
B CSM top-1 / MRR	0.6053 / 0.6870
C chemistry top-1 / MRR	0.6053 / 0.6870
D probabilistic top-1 / MRR	0.4027 / 0.5136
E Bayesian top-1 / MRR	0.6160 / 0.6911
C vs B delta top-1	+0.0000 95% CI [+0.0000, +0.0000]
C vs B McNemar p	1.0000
Model C weights chosen in-fold	alpha=0.4, beta=gamma=delta=0 in ALL five folds
Decompositions non-reconciling	0 of 120
Frozen inputs	LSM 208482d6... · CSM 0b4aa550... · engine 20d8bd99... — verified
Completed	2026-08-07 04:52:30 UTC

Sources of record: reports/PHASE_08_REPORT.md · PHASE_08_SCIENTIFIC_AUDIT.md · PHASE_08_DECISION_GATE.md
The nested weight search selected zero chemistry weight in every fold. Model C is therefore identical to Model B, and the difference between them is exactly zero by construction rather than by measurement error.

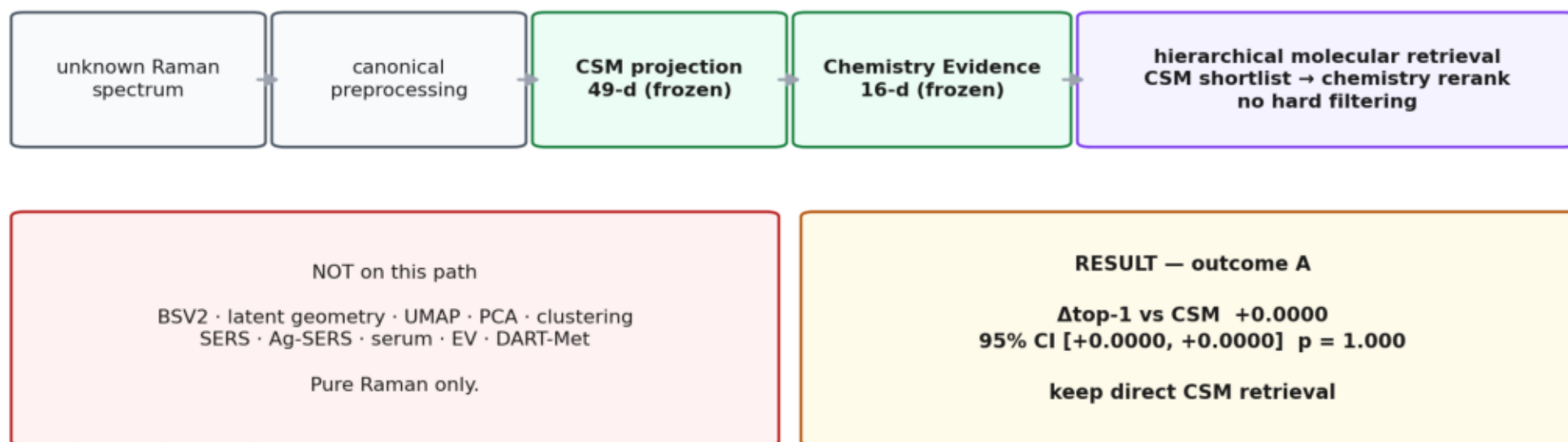
Figures

01 architecture
02 performance
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Figure 01

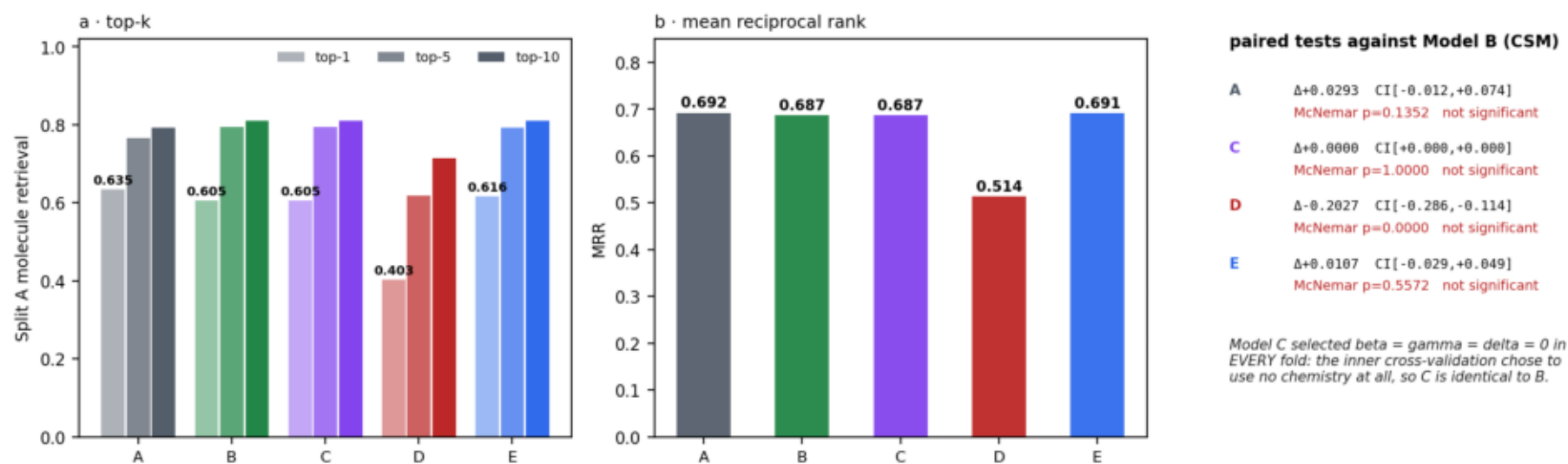
Figure 1 · Phase 08 architecture and canonical inference path



Phase 08 architecture and canonical inference path. BSV2 is not on it and is not imported anywhere in the retrieval package. The result box carries the paired test that decides the phase.

Figure 02

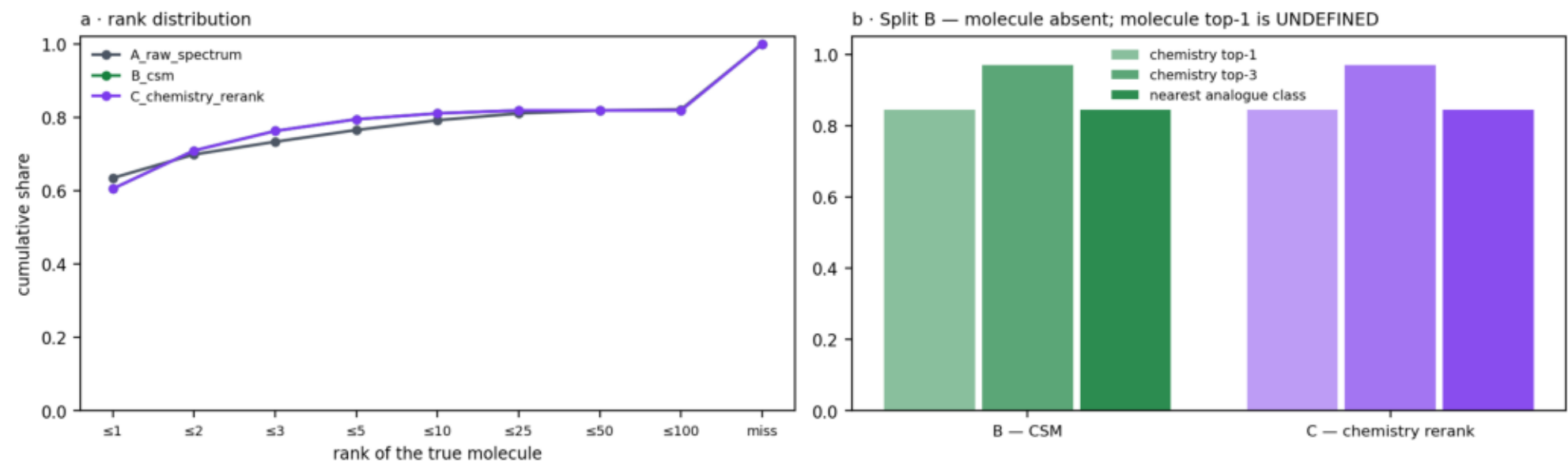
Figure 2 · Retrieval performance and significance — Split A, molecule present



Split A retrieval for all five models, with paired tests against Model B. Model C selected beta = gamma = delta = 0 in every fold: the inner cross-validation chose to use no chemistry at all, so C is identical to B and the difference is exactly zero.

Figure 03

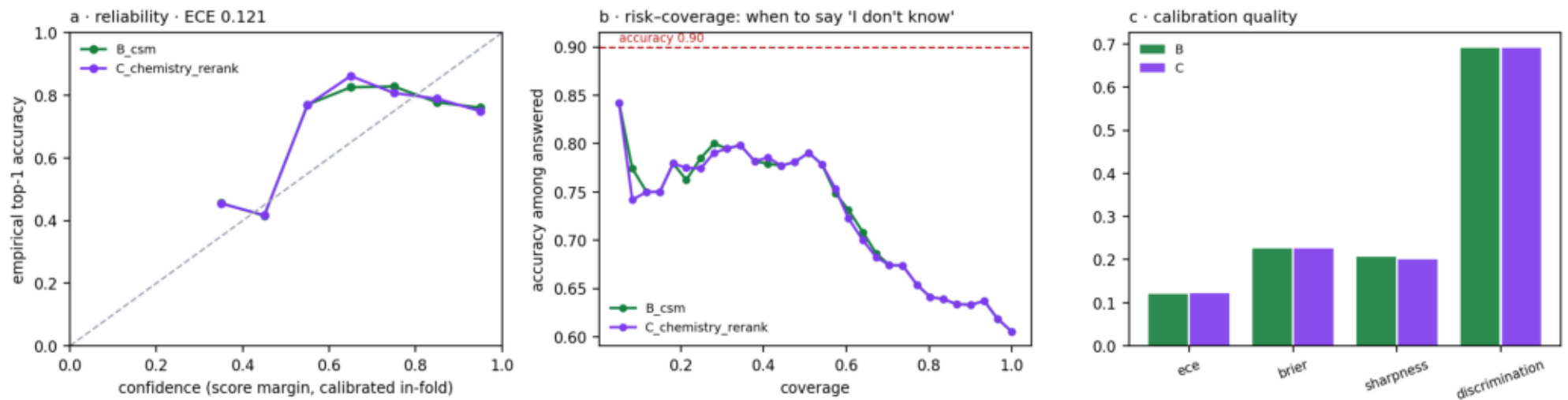
Figure 3 · Rank distribution and the held-out-molecule split



Rank distribution, and the held-out-molecule split where molecule top-1 is undefined rather than zero. Chemistry reranking changes neither.

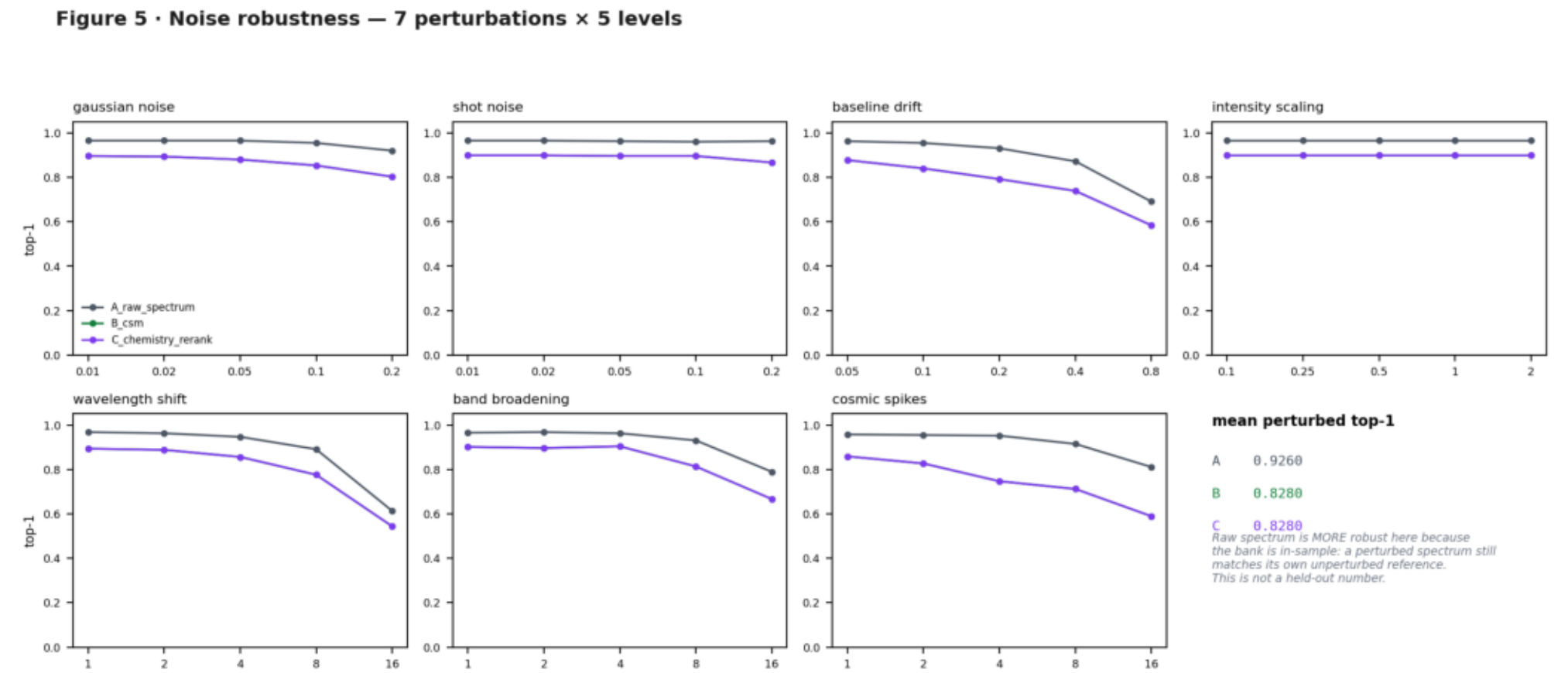
Figure 04

Figure 4 · Calibration, selective prediction and abstention



Calibration from the score margin, risk-coverage and abstention. An earlier version derived confidence from $1/\text{rank}$ -- a function of the quantity being predicted -- and scored a discrimination of exactly 1.000. That was circular, not good.

Figure 05

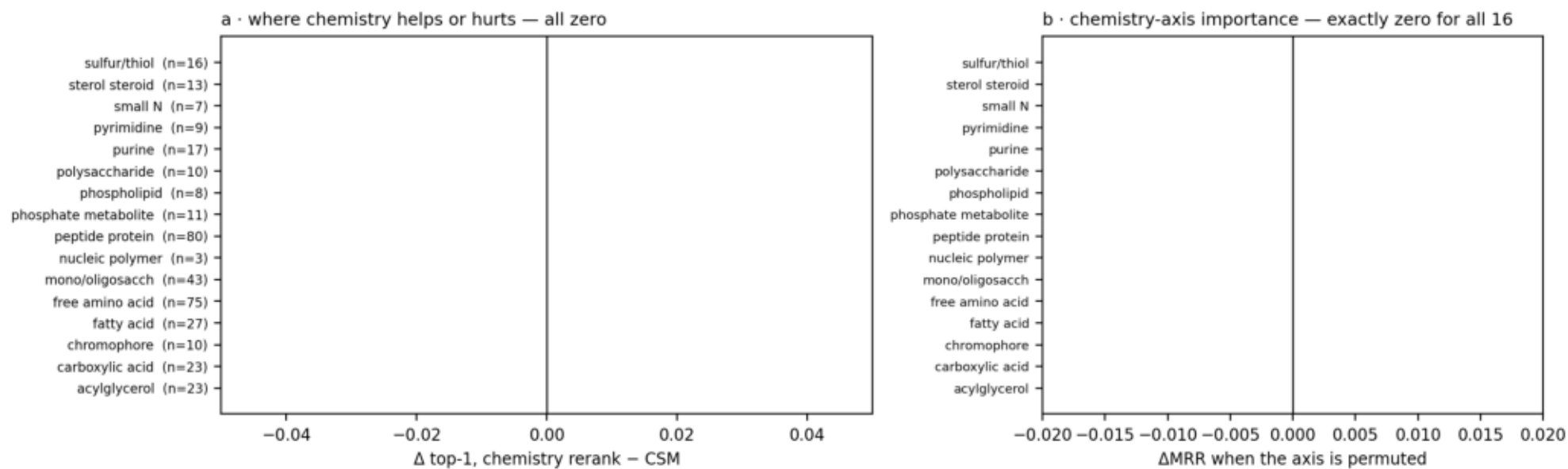


Noise robustness across seven perturbations. Raw spectrum appears most robust here because the bank is in-sample: a perturbed spectrum still matches its own unperturbed reference. It is not a held-out number.

Figure 06

Figure 6 · Failure analysis and chemistry-axis contribution

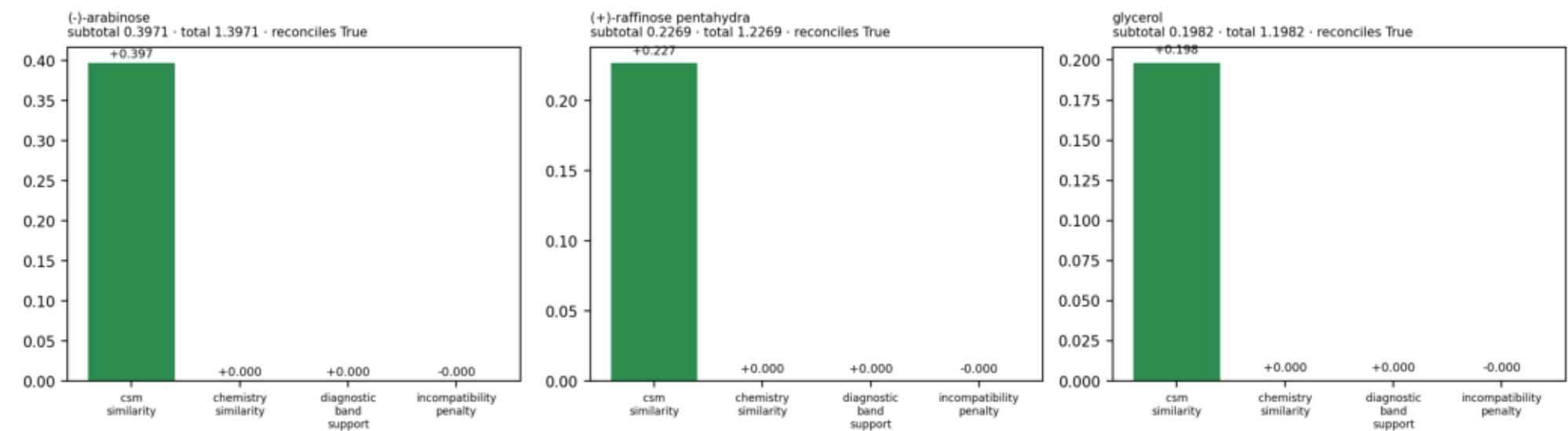
with $\beta = \gamma = \delta = 0$ selected in every fold, chemistry contributes nothing and both panels are flat by construction



Failure analysis and chemistry-axis permutation importance. Both are flat by construction: with zero chemistry weight there is nothing to help, hurt, or permute.

Figure 07

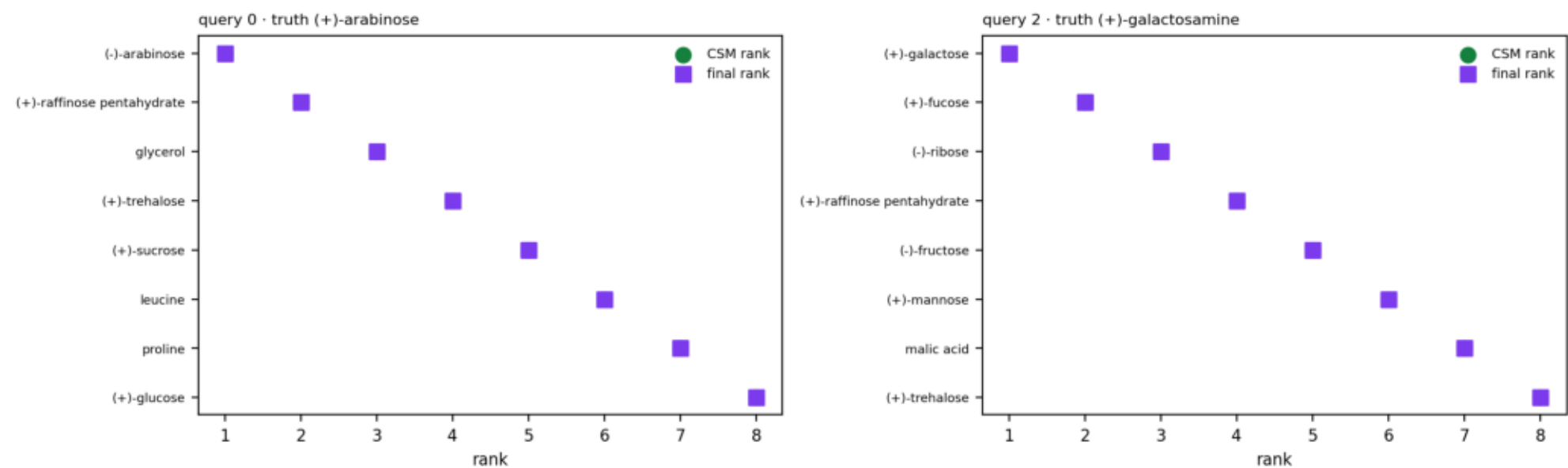
Figure 7 · Evidence decomposition — every score sums exactly, no hidden term
0 of 12 shown decompositions fail to reconcile



Evidence decomposition. Every Model C score is four weighted terms that sum to the displayed total, checked against the model's own output. Zero of 120 decompositions failed to reconcile.

Figure 08

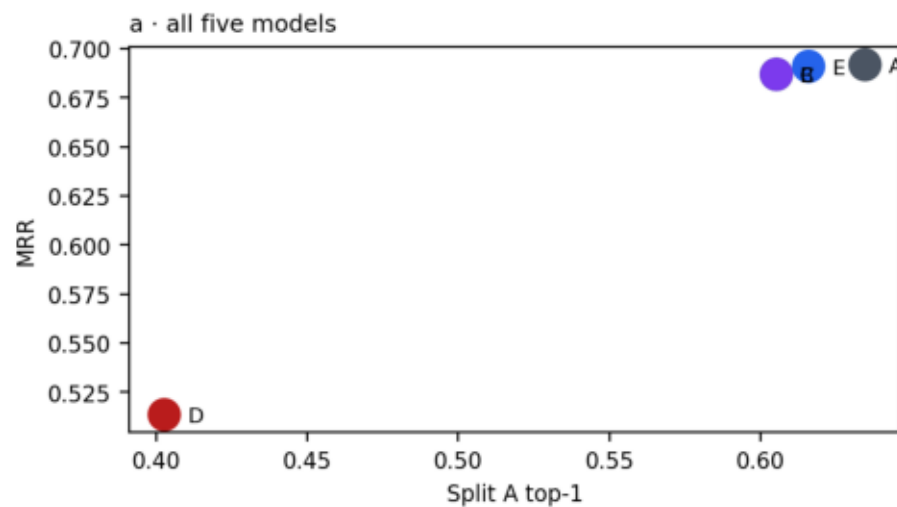
Figure 8 · Candidate evolution — CSM ranking versus final ranking
the two coincide exactly, because reranking selected zero chemistry weight



Candidate evolution from CSM ranking to final ranking. The two coincide exactly.

Figure 09

Figure 9 · Summary and decision gate



decision

baselines reproduced	EXACT
Model B (CSM) top-1	0.6053
Model C top-1	0.6053
Δ top-1	+0.0000
95% CI	[+0.0000, +0.0000]
McNemar p	1.0000
any metric significant	False
decompositions non-reconciling	0
OUTCOME	A

G1 frozen fingerprints verified	G2 baselines reproduced exactly before new work	G3 BSV2 absent from the inference path	G4 Raman-only scope	G5 all five models benchmarked	G6 Model C weights tuned in inner folds only
G7 no hard filtering — chemistry is a soft prior	G7b every model scored against an identical candidate bank	G7c confidence derived from the score margin, not from the rank	G8 every score decomposition reconciles	G9 Split B reports no molecule top-1	G10 improvement tested for significance, not asserted
G11 calibration and risk-coverage reported	G12 noise robustness across all three primary models	G13 failure analysis identifies where chemistry hurts	G14 chemistry-axis importance computed exactly	G15 deterministic on rerun	G16 decision follows the pre-declared A/B/C rule

18 of 18 gates pass · outcome A — keep direct CSM retrieval · BSV2 not on the inference path

Summary and decision gate. Outcome A: keep direct CSM retrieval.